

**B.Sc. (HONS.) IN CSE FIRST YEAR SECOND SEMESTER
EXAMINATION, 2021**

LINEAR ALGEBRA

[According to the New Syllabus]

Subject Code : 510225

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

Marks

1. (a) Define system of linear equation. Solve the following system of linear equation : 2+6=8

$$x_1 - x_2 + x_3 + x_4 = 0$$

$$x_1 + 2x_2 - x_3 - x_4 = 1$$

$$2x_1 - 2x_2 + x_3 - x_4 = 0$$

$$3x_2 - x_3 + x_4 = 1$$

- (b) Determine the value of λ such that the system of linear equation has 6

(i) No solution (ii) Many solution (iii) More than one solution
 $x + y - z = 1, 2x + 3y + \lambda z = 3, x + \lambda y + 3z = 2$

- (c) Solve the following system of linear equation by Matrix method. 6

$$2x + y - 2z = 10$$

$$3x + 2y + 3z = 1$$

$$5x + 4y + 3z = 4$$

2. (a) If $A = \begin{pmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{pmatrix}$ then prove that, $A^3 = A^{-1}$. 8

- (b) Prove that, $\begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = 2abc(a+b+c)^3$. 8

- (c) Show that every square matrix can be expressed as the sum of Hermitian and a Skew-Hermitian matrix. 4

[Please turn over]

Marks

3. (a) Define singular and nonsingular matrix. 2

(b) If A is a Skew Symmetric matrix then prove that, $AA^T = A^T A$ and A^2 is Symmetric. 8

(c) Define inverse Matrix. 2+8=10

← If $A = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$. Find $A^{-1} B$.

4. (a) Define vectors in $\mathbb{R}^n$. Prove that, $|u \cdot v| \leq \|u\| \cdot \|v\|$, $\forall u, v \in \mathbb{R}^n$. 8

(b) Define Norm in C^n . If $v, u \in \mathbb{R}^2$ where $u = (5, -2)$ and $v = (3, 4)$. Then prove that, $\|u + v\| \leq \|u\| + \|v\|$. 6

(c) What is unit vector in $\mathbb{R}^n$? Let $u, v, w \in \mathbb{R}^3$ where $u = (1, a, 2, -1)$ $v = (2, 1, 3, b)$ and $w = (b, 1, -4, 2)$. If the vector u perpendicular to v and w , then find the value of a and b . 2+4=6

5. (a) Define kernel and image of a linear transformation. A linear operator $T : u \longrightarrow u$ on a vector space of finite dimension is invertible, then prove that it is nonsingular. 4+4=8

(b) Let S and T be the linear transformations $S, T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ defined by $S(x, y, z) = (y, z, x)$ and $T(x, y, z) = (x + y + z, 0, 0)$. Find basis for the kernel of $S + T$. 6

(c) Find standard matrix for the linear transformation $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^4$ defined by $T(x, y, z) = (x + y, x - y, z, x)$. 6

6. (a) Define characteristic polynomial and characteristic equation. 4

(b) State Cayley-Hamilton theorem. For the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ 2+6=8

Prove that, $A^3 - A^2 - 15A - 15I = 0$.

(c) Find the eigen values and corresponding eigen vectors of the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$. 8